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Burstiness & Perplexity | Definition & Examples

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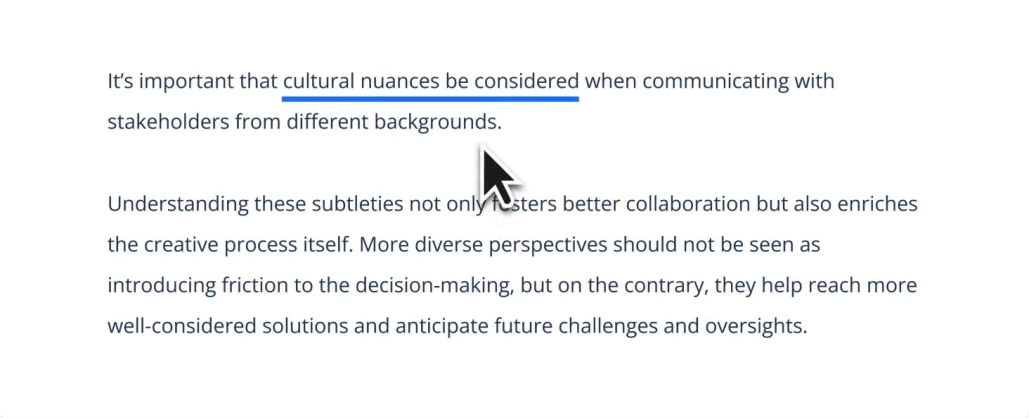
Perplexity and burstiness are two important metrics used to distinguish human writing from AI-generated text.

- **Perplexity** refers to how predictable or unpredictable a piece of text is (or how "surprised" a model is by the word choices), with higher perplexity often linked to more complex, human-like writing.
- **Burstiness** focuses on variation in sentence length and structure. Higher burstiness, with a mix of short and long sentences, tends to feel more natural and human, while more uniform and repetitive patterns are often a sign of AI-generated writing.

AI detectors like QuillBot's AI Detector can use these metrics to quantify how human-like a piece of writing is, while QuillBot's Humanize AI tool can help refine AI-generated text.



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What are burstiness and perplexity?

"Perplexity" and "burstiness" describe two components of a piece of writing: how predictable it is (perplexity) and how varied the overall structure is (burstiness).

Perplexity definition

Perplexity measures predictability—how easy it is to guess what comes next.

Consider the sentence "I went for a hike in..." A predictable, or low-perplexity, outcome might be something like "the woods." "Muskoka" is also a valid ending to this sentence, but it's harder to predict, so it would have moderate perplexity. An even more unlikely, high-perplexity answer would be "a distant galaxy inhabited solely by sea sponges."

Errors like typos and misspellings occur more frequently in human-written text than in AI-written text, and because they are unpredictable, they also increase perplexity.

The idea of predictability is core to how AI writers work. These large language models (LLMs) are trained on massive amounts of text, and they use this training to predict which words should come next. As a result, AI-written text tends to have lower perplexity than human-written text.

Burstiness definition

While perplexity considers how predictable text is at the level of individual words, **burstiness** considers the bigger picture—how much variation exists between sentences and throughout the overall text.

Burstiness measures how factors like sentence length, sentence structure, and word use shift throughout a document. People tend to write and speak in long and short bursts—we alternate between short, punchy sentences and more complex ones. Generative AI, on the other hand, will often use more uniform, repetitive sentence structure. Its burstiness is therefore generally lower than that of a human.

Consider the two examples, and see if you can guess which one is higher in burstiness.

High and low burstiness example

Example 1: *Today I woke up at 7 a.m. I walked to the kitchen. I had breakfast and did a crossword. I turned on my computer at 8 a.m. I worked until noon. I had pasta salad and some Greek yogurt for lunch. In the afternoon I did more work. After work I went to trivia with my friends.*

Example 2: *I woke this morning at 7 a.m. to the sound of my blaring alarm. Reluctantly, I stumbled to the kitchen. The usual business of breakfast and slowly resurfacing to consciousness was considerably improved by my daily crossword; today's was lightly challenging with some clever clues. Work, which dragged on from 8 a.m. until noon, was much less engaging, and lunch comprised an uninspiring repast of pasta salad and Greek yogurt. My afternoon brought more of the same—four more hours of uninterrupted work. But fear not! The tedium finally broke in the evening, when I headed to my favorite pub for trivia with some friends.*

Although both excerpts provide the same information, you probably noticed a big difference between the two. If you

guessed that Example 2 has higher burstiness, you're absolutely right! Its sentence structure varies considerably compared to the flat and repetitive Example 1.

QuillBot's Humanize AI can help vary sentence length, structure, and rhythm so your text feels more natural, dynamic, and human-like.

Burstiness vs perplexity

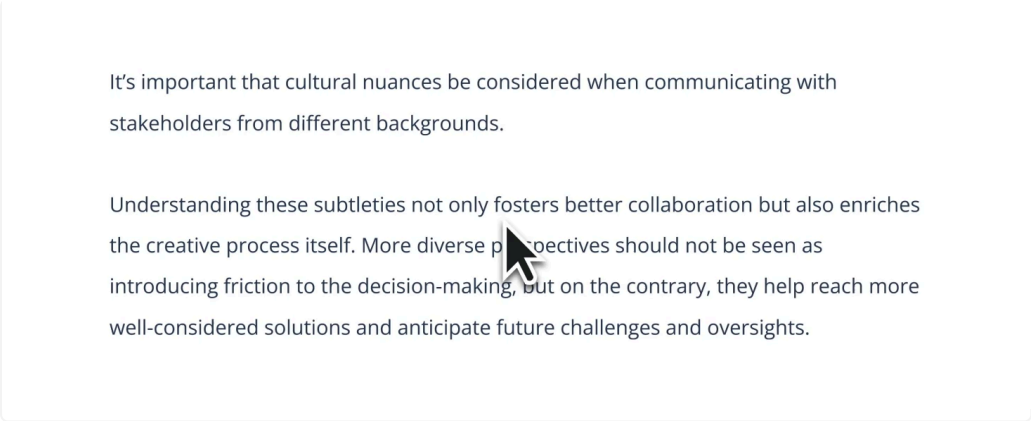
Burstiness and perplexity can both be used to assess whether a piece of text appears more AI-like or human-like, but they consider separate elements of writing. Perplexity measures predictability at the level of words, whereas burstiness measures predictability at the level of sentences and text structure.

For instance, a text with low burstiness might contain sentences that are all the same length and style, whereas a text with low perplexity would use highly predictable words. Consider the following examples of how different levels of burstiness and perplexity can change how a piece of writing sounds.

Examples of perplexity vs burstiness

	Low Burstiness	High Burstiness
Low Perplexity	I went to a new restaurant for dinner tonight. I sat down. I ordered a burger. I ate it. I paid my bill. I went home.	I decided to try a new restaurant tonight. The hostess led me to a booth by the window, where I scanned the menu, settled on a burger, and ordered. Ten minutes later it arrived—fries golden and crisp, bun perfectly toasted. After eating, I stayed for a coffee before heading home.
High Perplexity	I went to a bookstore for dinner tonight. I sat down. I ordered deep-	Forty-seven days ago, I went to a library for dinner. Strange? Absolutely. The hostess—a stern librarian with horn-rimmed glasses—seated me in a fireside armchair

	Low Burstiness	High Burstiness
	fried ice cream. I massaged it into my hair. I ate my bill. I crabwalked home.	between the poetry and philosophy sections. I ordered the alphabetized soup. It arrived steaming, with a side of Catcher in the Rye toast. Edible literature! When I chuckled, the diners nearby shushed me with theatrical fury. I whispered an apology, borrowed the meal using my library card, and tiptoed out feeling surprisingly well-read.



Free AI Detector

Burstiness, perplexity, and AI detection

Because burstiness and perplexity measure how human-like writing is, they're a great way to evaluate AI text generators. Developers and researchers can use them to evaluate and enhance model outputs.

These measures can also capture certain styles of writing. For example, an academic paper—which has distinct sections, each with unique structure and terminology—has higher burstiness than a poem, which usually follows a more predictable rhythm. Recognizing these differences can help models generate more realistic text.

Burstiness and perplexity can also be used to determine whether a piece of writing is AI-generated. AI detection tools will combine these measures with other indicators to determine the likelihood that

something was written by an AI. However, as AI text generators become increasingly sophisticated, distinguishing human from AI will become a much harder task.

Frequently asked questions about burstiness and perplexity

- What is an example of burstiness?
- What is an example of perplexity?
- What are burstiness and perplexity used for?
- How do I check if a document is AI generated?
- Are higher perplexity and higher burstiness always better?

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Emily has a bachelor's degree in electrical engineering, a master's degree in psychology, and a PhD in computational neuroscience. Her areas of expertise include data analysis and research methods.

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